ASHWIN DEWAN

Embedded Software Developer

zeakashwin@gmail.com 7042586242

Embedded Software Developer with hands-on experience in Embedded C/C++ and Linux-based systems, working across wireless devices, IoT products, and hardware-integrated firmware. Experienced in debugging, protocol integration, and performance optimization for production embedded systems. Strong background in microcontroller programming, device communication, and system-level problem solving.

Skills

Embedded & Systems

- Microcontroller Programming & Firmware Development
- Linux Embedded Systems
- Real-Time Operating Systems (RTOS)
- Device Drivers & Low-Level Hardware Interaction
- Cross Compilation & Build Systems
- Yocto

Debugging

- Low-Level Debugging & Root Cause Analysis
- Hardware–Software Troubleshooting
- System Performance Optimization
- Memory & Resource Optimization

Communication & Networking

- I2C
- SPI
- UART
- USB
- BLE / Bluetooth
- TCP/IP
- UDP
- HTTPS
- Wi-Fi

Tools

- Gdb & Strace
- Serial Debugger
- STM32 Cube IDE
- Visual Studio Code
- Putty (SSH remote connection)
- SVN & Git (version control)
- Android Studio
- WinSCP (Remote file manager)
- GCC & G++ compilers
- YOLO (You Only Look Once)

Work Experience

Embedded Developer at Qubo Innovation

Sep 2024- Current

- **Multi-Device Firmware Development:** Developing and maintaining embedded firmware across wireless and IoT devices, including security cameras, smart locks, air purifiers, and a BLE hub.
- **Embedded System Logic & Control:** Implementing device control logic, hardware interaction, and system behavior for reliable real-world operation.
- **Wireless & Device Communication:** Designing and debugging wireless connectivity and peripheral communication mechanisms, including BLE-based device interactions.
- **Low-Level Debugging & Analysis:** Performing fault isolation, root cause analysis, and low-level debugging across hardware–software boundaries.
- **Performance & Resource Optimization:** Optimizing firmware performance, memory usage, and system responsiveness for production devices.
- **Feature Development & Enhancements:** Supporting new feature implementation, system improvements, and bug fixes across embedded product lines.
- **Cross-Functional Collaboration:** Working closely with hardware and application teams to ensure seamless device integration and functionality.

Embedded Developer at Safety Labs *July 2022- June 2024*

- **Linux-Based Embedded Development:** Developed and maintained embedded applications on Linux-based hardware platforms, focusing on system behavior and reliability.
- **Core Platform & System Engineering:** Contributed to core software architecture and platform components supporting multiple embedded products.
- **Hardware & Peripheral Integration:** Enabled communication with hardware peripherals through low-level interfaces and device interactions.
- **System-Level Problem Solving:** Diagnosed and resolved system-level issues affecting performance, stability, and device functionality.
- **Structured Development & Version Control**: Participated in collaborative development workflows using Git/SVN within controlled release cycles.

Junior C/C++ Developer at VCSE (Part-Time) *Nov 2018 – Mar 2022*

- **C/C++ Programming & Foundations:** Provided technical guidance on C/C++ programming, focusing on core concepts, memory handling, and program structure.
- **Applied Problem Solving:** Assisted in designing and implementing solutions for algorithmic and logic-building challenges.
- **Project & Implementation Support**: Supported development of academic and practical projects involving software logic and system behavior.
- **Code Quality & Best Practices**: Helped improve code organization, readability, and structured development approaches.
- **Technical Knowledge Sharing:** Conducted sessions on advanced programming topics and practical development techniques.

Projects

In Room Tv

- Developed embedded applications for TV-based user interaction and communication features.
- Implemented device-side logic for personalized profiles, messaging, and accessibility workflows.
- Worked with Linux-based hardware platforms for system behavior and feature integration.
- Contributed to interface responsiveness and system stability for continuous operation

Telehealth Kiosk

- Developed embedded logic supporting secure access and device interaction workflows.
- Implemented monitoring and notification-related system behavior.
- Optimized runtime stability for continuous device operation.
- Participated in debugging and system refinement across hardware-integrated components.

Telehealth Cart

- Contributed to embedded software development for a point-of-care telehealth system.
- Supported device functionality related to video workflows and peripheral integrations.
- Worked on Linux-based system behavior and hardware interaction.
- Assisted in improving system reliability for real-time medical usage scenarios.

Smart Traffic Lights

- Developed logic for traffic-aware signal control using computer vision outputs.
- Integrated detection results into dynamic timing and control decisions.
- Focused on system behavior and automation reliability.
- Contributed to improving responsiveness of traffic control workflows.

Education

Bachelors in technology | EEE

BHAGWAN PARSHURAM INSTITUTE OF TECHNOLOGY *Aug 2018 - July 2022*
Acquired 75% in Btech

High School & Senior Secondary School *April 2014 - May 2018*

Acquired 80 % in class 12TH
Acquired 85 % in class 10TH